

Companion XLVIII: Minkowski Functionals in the Relaxation-Driven Cyclic Cosmology

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Abstract

Minkowski functionals provide a complete morphological and topological description of random fields and have become a powerful tool for analyzing the large-scale structure of the Universe. In the standard cosmological model, the topology of the density field is determined by Gaussian initial conditions and non-linear gravitational evolution. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the morphology, connectivity, and topology of the cosmic web. This Companion develops a continuous description of Minkowski functionals in RDCC, deriving how the relaxation scale influences the volume fraction, surface area, mean curvature, and Euler characteristic of the density field. The resulting framework provides a distinctive topological signature of the RDCC gravitational field.

1 Introduction

The large-scale structure of the Universe is not only characterized by its power spectrum or bispectrum but also by its topology and morphology. Minkowski functionals provide a complete set of geometric descriptors that quantify the shape, connectivity, and curvature of isodensity surfaces. They capture information beyond correlation functions and are sensitive to non-Gaussianity, mode coupling, and the global structure of the cosmic web.

In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the topology of the density field in a scale-dependent manner, producing distinctive signatures in the Minkowski functionals. These signatures provide a direct probe of the relaxation scale k_{rel} .

2 Minkowski Functionals of the Density Field

For a three-dimensional field, the Minkowski functionals are defined as:

$$V_0(\nu) = \text{Volume fraction}, \tag{1}$$

$$V_1(\nu) = \text{Surface area}, \tag{2}$$

$$V_2(\nu) = \text{Integrated mean curvature}, \tag{3}$$

$$V_3(\nu) = \text{Euler characteristic}. \tag{4}$$

Here ν is the density threshold in units of the standard deviation.

In the standard cosmological model, these functionals follow analytic expressions for Gaussian random fields, with deviations arising from non-linear evolution.

In RDCC, the scale-dependent evolution of the gravitational potential modifies the density field in a way that alters all four functionals.

3 RDCC Modifications to the Density Field

The density contrast evolves according to

$$\delta_{\text{RDCC}}(k, a) = \delta(k, a) \left[1 - \chi_\delta \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (5)$$

This modification suppresses small-scale fluctuations and enhances large-scale coherence. The resulting density field is smoother, with fewer small-scale peaks and a more coherent filamentary structure.

These changes propagate directly into the Minkowski functionals.

4 RDCC Minkowski Functionals

The Minkowski functionals of a weakly non-Gaussian field can be written as

$$V_i(\nu) = V_i^{(G)}(\nu) + \Delta V_i(\nu), \quad (6)$$

where $V_i^{(G)}$ is the Gaussian prediction and ΔV_i encodes non-Gaussian corrections.

In RDCC, the corrections become

$$\Delta V_{i,\text{RDCC}}(\nu) = \Delta V_i(\nu) \left[1 - \chi_{V_i} \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (7)$$

This modification produces:

- a smoother volume fraction curve,
- reduced surface area at high thresholds,
- enhanced mean curvature at intermediate thresholds,
- and a distinctive shift in the Euler characteristic.

The Euler characteristic, in particular, becomes a sensitive probe of the relaxation scale.

5 Topology of the Cosmic Web in RDCC

The Euler characteristic counts the number of connected components minus the number of tunnels plus the number of cavities. In RDCC, the suppression of small-scale modes reduces the number of isolated peaks and tunnels, while the enhanced coherence of large-scale modes increases the size and connectivity of filaments.

The topology of the cosmic web becomes smoother and more coherent, with fewer small-scale structures and more prominent large-scale features.

This topological signature provides a direct observational probe of the RDCC gravitational field.

6 Conclusion

Minkowski functionals in the Relaxation-Driven Cyclic Cosmology reflect the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the morphology, connectivity, and topology of the cosmic density field, producing distinctive signatures in the volume fraction, surface area, mean curvature, and Euler characteristic. These effects provide a powerful topological probe of the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of cosmic topology, non-Gaussianity, and the geometry of large-scale structure in RDCC.

References

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